



**K.RAMAKRISHNAN
COLLEGE OF ENGINEERING**

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

IBM FINAL PROJECT REPORT

END TO END DATA ANALYTICS APPLICATIONS

SUBMITTED BY

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3.JELSIA R



DECLARATION

We hereby declare that the project titled “End to end data analytics applications” is a bonafide work carried out by us.

This project report is submitted in partial fulfillment of the requirements for the completion of the course “Python for Data Science.”

We further declare that this work has not been submitted to any other institution or university for the award of any degree or diploma. The information presented in this report is true to the best of our knowledge and belief.

LAKSHANA SRI . B

JENYA . D

JELSIA. R

Place: K.RAMAKRISHNAN COLLEGE OF ENGINEERING

Date: 13/04/2026



BONAFIDE CERTIFICATE

This is to certify that the project titled “END TO END DATA ANALYTICS APPLICATIONS” is the bonafide work done by **LAKSHANA SRI. B, JENYA. D,** and **JELSIA. R** hereby and first-year students of K.Ramakrishnan college of engineering, Samayapuram, Trichy, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Artificial Intelligence and data science 2025-2026.

PROJECT GUIDE:

HEAD OF THE DEPARTMENT:

DATE: 13/04/2026



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ABSTRACT

The Music Listening Habits Analytics System is an end-to-end data analytics application designed to analyze users' music consumption patterns. The system enables users to upload raw listening data, which is then cleaned, processed, and analyzed to generate meaningful insights. Visualization techniques are used to present trends such as favorite genres, most played artists, listening time patterns, and mood-based preferences. This project demonstrates how data analytics can enhance user experience and support recommendation systems in music streaming platforms.



INTRODUCTION

With the rapid growth of digital music platforms like Spotify and Apple Music, massive amounts of user listening data are generated daily. Analyzing this data helps in understanding user behavior and preferences. This project focuses on building a pipeline that transforms raw listening data into actionable insights through data analytics techniques.



PROBLEM STATEMENT

Music streaming platforms collect large volumes of user data, but raw data alone does not provide meaningful insights.

There is a need for a system that:

- Cleans and processes raw listening data
- Identifies listening patterns and preferences
- Visualizes insights effectively
- Supports personalized recommendations.



OBJECTIVE OF THE PROJECT

Main Goal:

To develop an end-to-end analytics system that extracts meaningful insights from music listening data.

Specific Objectives:

- To collect and upload music listening datasets
- To clean and preprocess raw data
- To analyze listening patterns
- To create interactive visualizations
- To derive user-centric insight



SCOPE OF THE PROJECT

1. Personal listening behavior analysis
2. Genre and artist popularity trends
3. Time-based listening patterns
4. Integration with recommendation systems
5. Useful for streaming platforms, researchers, and marketers



LITERATURE REVIEW & EXISTING SYSTEM

Literature Review:

Previous studies in **music analytics**
and **user behavior analysis** show that:

1. Data mining techniques improve recommendation systems
2. Machine learning models enhance personalization
3. Visualization improves interpretability



Existing System:

- I. Platforms like Spotify and YouTube Music provide basic analytics (e.g., yearly summaries)
- II. Limited customization and deep analytics for users
- III. Lack of transparency in recommendation algorithms



PROPOSED STATEMENT

The proposed system provides:

1. Full data pipeline (Upload → Clean → Analyze → Visualize)
2. Advanced analytics (trend detection, clustering)
3. Custom dashboards for users
4. Interactive visualizations
5. Better insights compared to existing systems



DATASET DESCRIPTION

This datasets typically include :

- 1.User ID
- 2.Song name
- 3.Artist name
- 4.Genre
- 5.Timestamp (date & time)
- 6.Duration of listening
- 7.Device/platform used

Data Sources:

- 1.Exported user data from Spotify
- 2.Simulated datasets (CSV format)
- 3.Public datasets (e.g., Kaggle)



TOOLS AND TECHNOLOGY USED

Data processing libraries :

1. Pandas
2. Numpy

Data Visualization

Web applications framework :

1. Streamlit

Data source

Development environment:

1. Visual studio code



METHODOLOGY

Step 1: Data Upload

Import dataset (CSV/Excel)

Validate file structure

Step 2: Data Cleaning

Remove duplicates

Handle missing values

Convert timestamps

Standardize genres and artist names



Step 3: Data Analysis

Most played songs/artists

Genre distribution

Listening frequency

Time-based trends (hour/day/month)

Step 4: Data Visualization

Bar charts (Top artists)

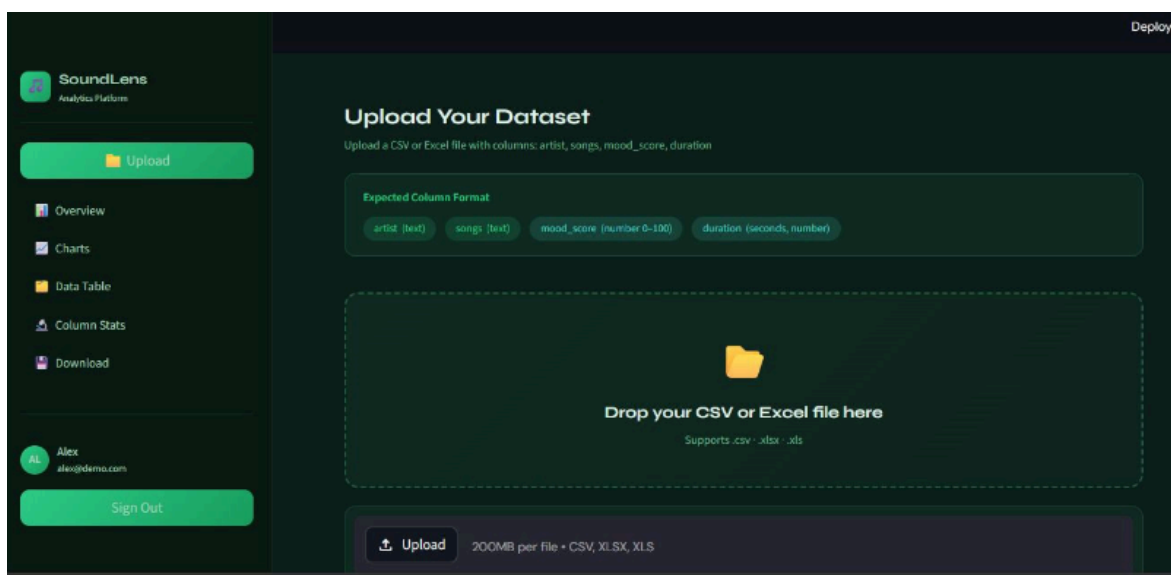
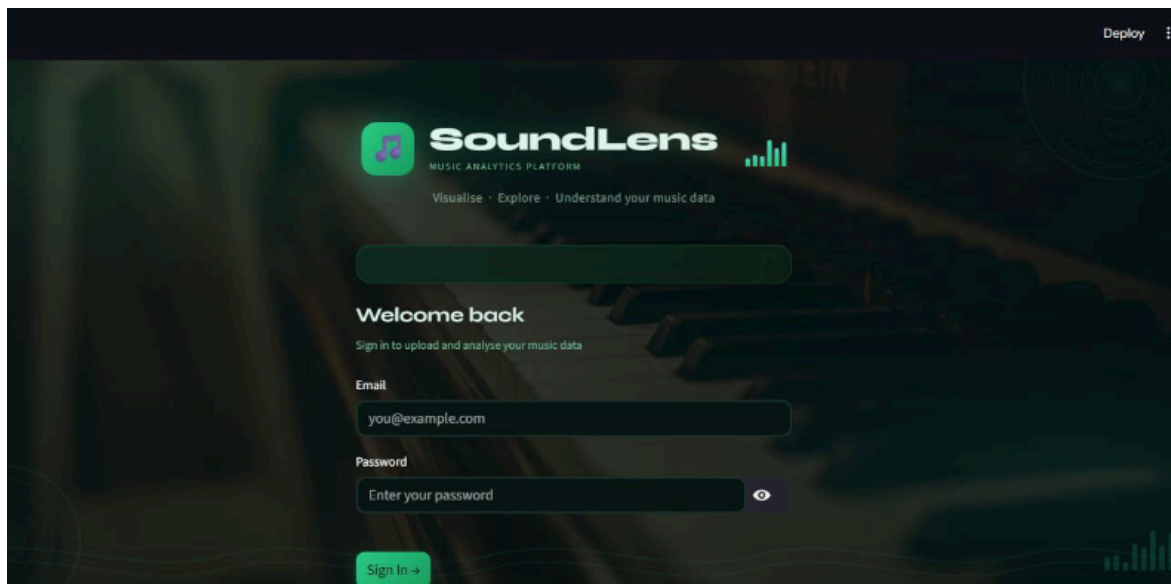
Pie charts (Genre share)

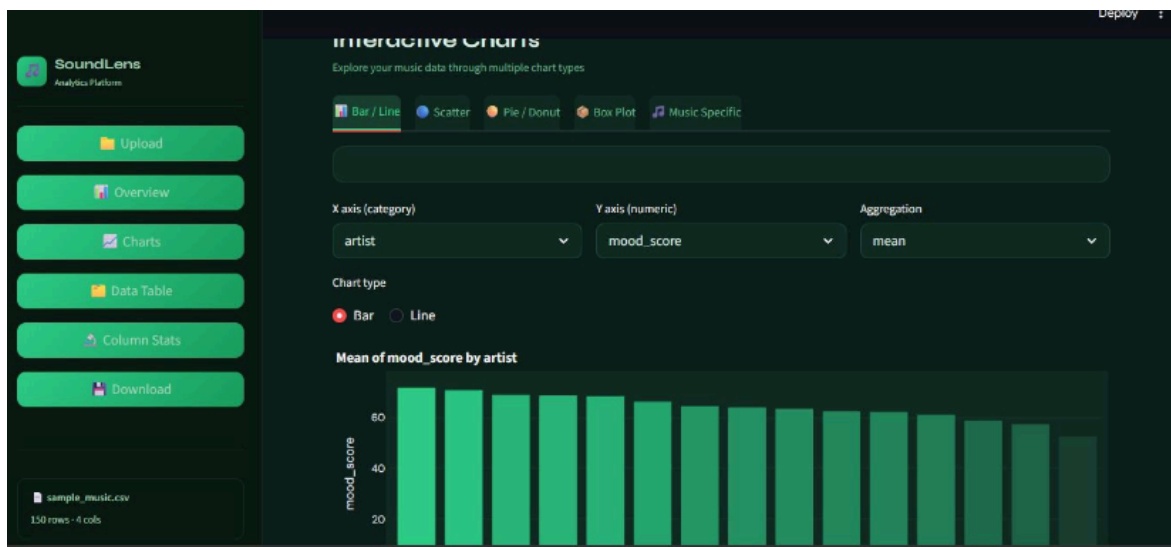
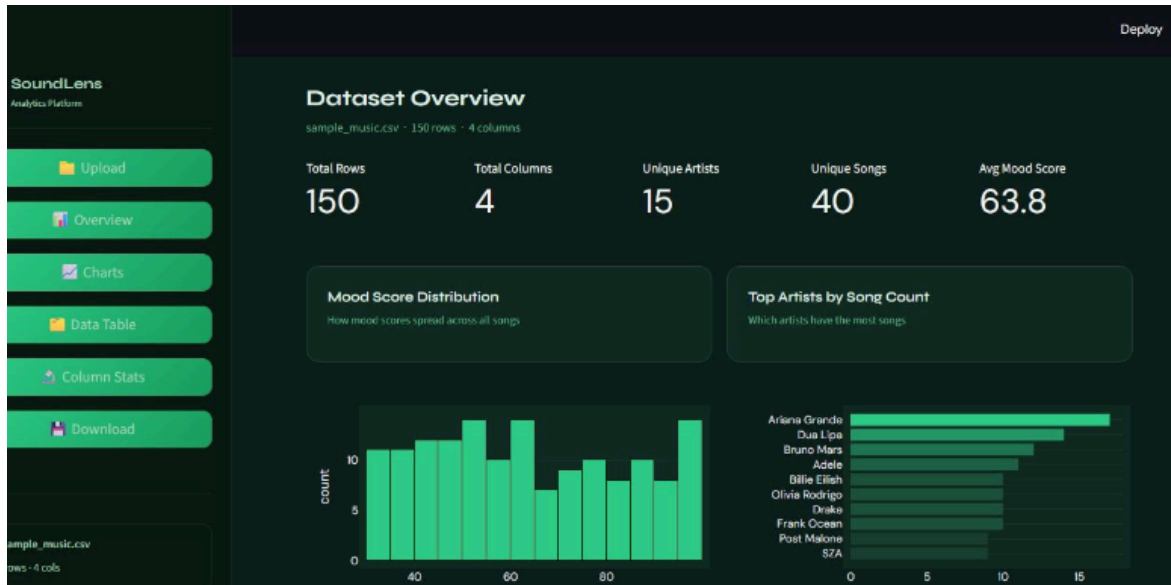
Line graphs (Listening trends)

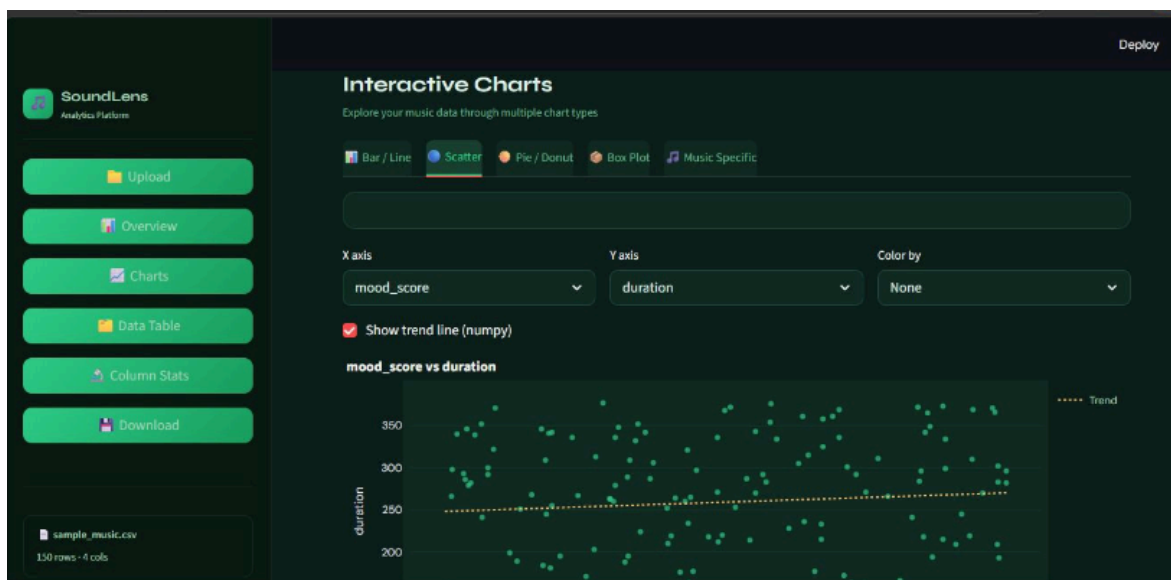
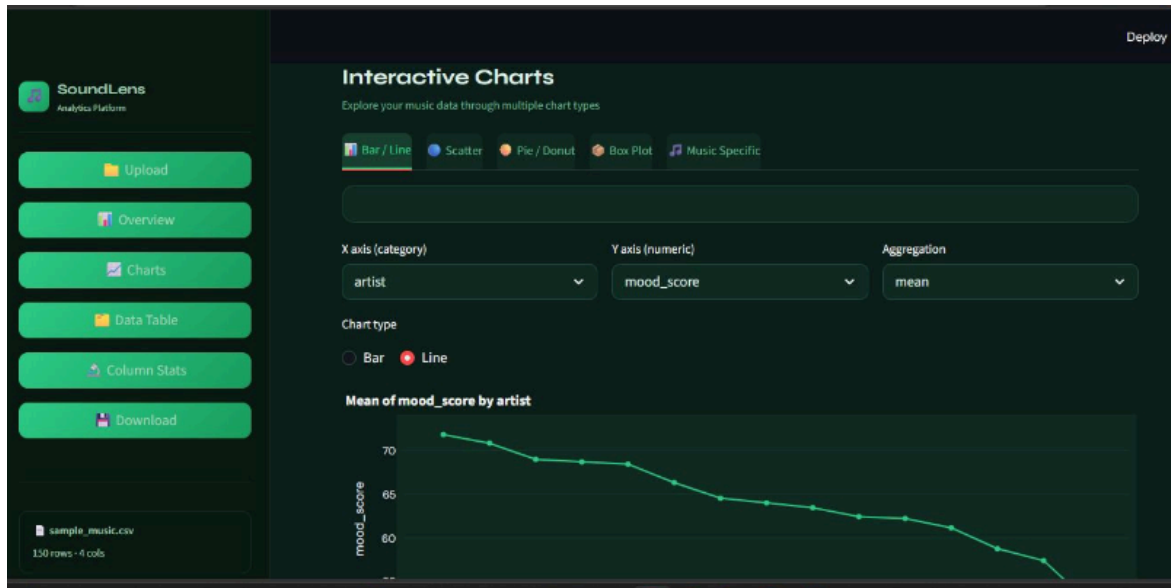
Heatmaps (Time vs activity)

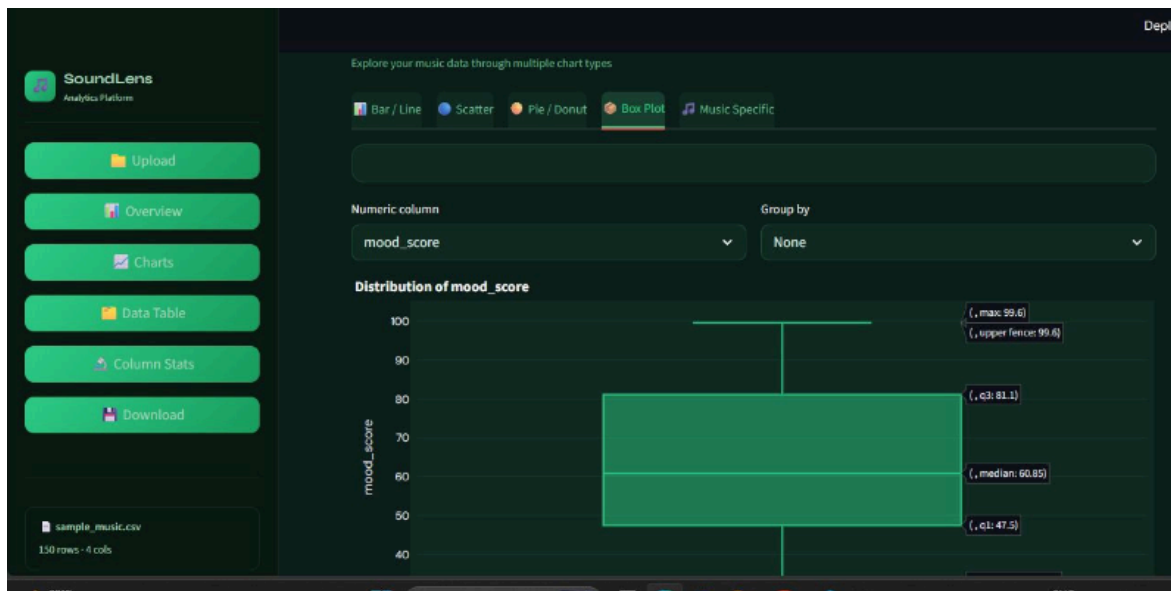
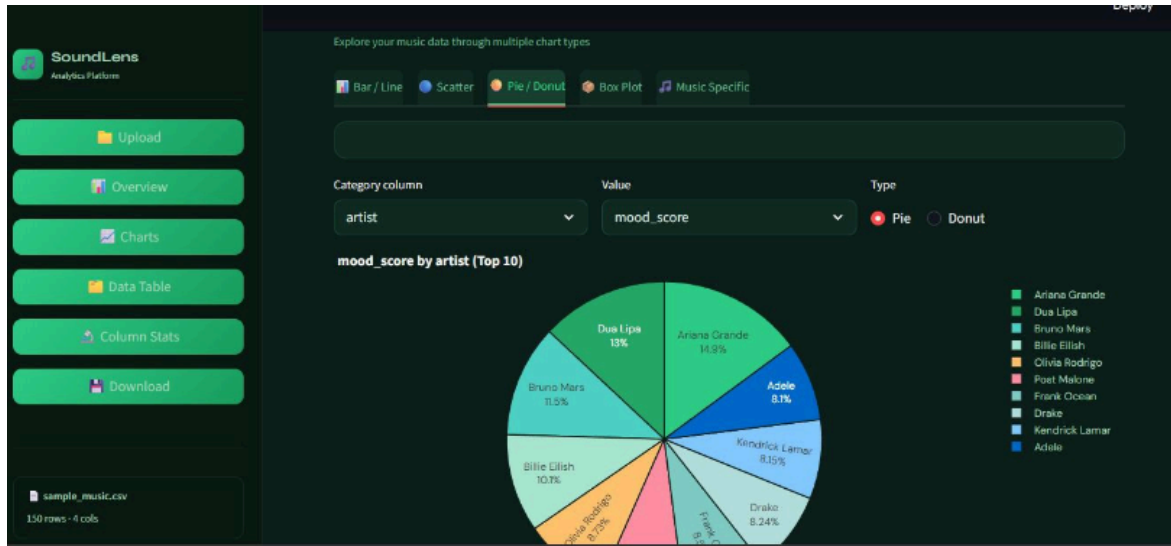


IMPLEMENTATION(SAMPLE CODE)&OUTPUT SCREEN SHOTS











SoundLens
Analytics Platform

Upload

Overview

Charts

Data Table

Column Stats

Download

sample_music.csv
150 rows · 4 cols

Deploy

Data Table

Browse, filter and search your raw data

Search value

Type to filter rows...

Filter column

All Columns

Rows to show

50

Showing 10 of 150 rows

	artist	songs	mood_score	duration
0	Post Malone	Out of the Woods	91.8	373
1	SZA	Welcome to New York	43.2	341
2	Kendrick Lamar	Bejeweled	49.5	142
3	Bruno Mars	Heat Waves	79	369
4	Ed Sheeran	Peaches	89.3	129
5	Dua Lipa	Stay	89.9	365
6	Kendrick Lamar	Shivers	58.3	219

SoundLens
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Upload

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sample_music.csv
150 rows · 4 cols

Deploy

Column Statistics

Deep-dive stats for any column

Select a column

artist

Total Values

150

Unique Values

15

Missing

0

Missing %

0.0%

Column type str — basic info shown above.



PERFORMANCE

- *Efficient processing of large datasets
- *Faster query execution using optimized data structures
- *Scalable for increasing data size

EVALUATION

- *User satisfaction
- *Comparison with existing platform analytics



CHALLENGES FACED

1. Handling missing or inconsistent data
2. Data normalization issues
3. Large dataset processing
4. Visualization clarity for complex data
5. Integration of multiple tools



FUTURE ENHANCEMENTS

- >Integration of Machine Learning for recommendations
- >Real-time data analytics
- >Mobile app dashboard
- >Sentiment analysis of songs
- >AI-based playlist generation



CONCLUSION

The Music Listening Habits Analytics System successfully demonstrates an end-to-end data analytics pipeline. It transforms raw listening data into meaningful insights using cleaning, analysis, and visualization techniques. The system enhances understanding of user behavior and can be extended to support advanced recommendation systems in modern music platforms.



REFERENCES

- 1.Kaggle
- 2.python for data analysis
- 3.GeeksforGeeks